

Nmap Vulnerability Assessment Report

Kali Linux Virtual Machine

1. Summary

A vulnerability and network exposure assessment was conducted against the Kali Linux virtual machine using Nmap. The assessment first identified the VM's network configuration and active hosts on the 192.168.64.0/24 subnet, followed by TCP service enumeration, an Nmap vulnerability-script scan, and a UDP scan of the 100 most common UDP ports. The results show that the Kali VM was reachable but did not expose any open TCP ports in the full 1–65,535 TCP port scan, and the 100 UDP ports tested were reported as closed. The vulnerability-script scan therefore identified no network-service vulnerabilities in the tested TCP ports.

2. Assessment Environment

Item	Observed Configuration
Operating system / assessment host	Kali Linux virtual machine
Network interface	eth0
Kali IPv4 address	192.168.64.2/24
Subnet	192.168.64.0/24
Broadcast address	192.168.64.255
Default gateway	192.168.64.1
Scanner	Nmap 7.94

3. Methodology

- 1. Network configuration:** The Kali VM's IPv4 address, subnet, interface, and gateway were identified using `ip addr`.
- 2. Asset discovery:** Nmap host discovery (`-sn`) was used against 192.168.64.0/24 to identify active systems.
- 3. TCP service enumeration:** Nmap service/version detection (`-sV`) with `-p-` was used to examine all 65,535 TCP ports on 192.168.64.2.
- 4. Vulnerability assessment:** Nmap's vulnerability-oriented NSE scripts were run with service detection against the Kali VM.
- 5. UDP assessment:** Nmap scanned the 100 most common UDP ports to identify potentially exposed UDP services.

4. Scan Configuration

```
ip addr
```

```
nmap -sn 192.168.64.0/24
```

```
nmap -sV -p- 192.168.64.2
```

```
nmap -sV --script vuln 192.168.64.2 -oN kali-vulnerability-scan.txt
```

```
nmap -sU --top-ports 100 192.168.64.2
```

5. Scan Results

5.1 Asset Discovery

The host-discovery scan examined all 256 addresses in the 192.168.64.0/24 subnet and identified two hosts as up: 192.168.64.1 and 192.168.64.2. The Kali VM is confirmed as 192.168.64.2. The 192.168.64.1 address corresponds to the gateway observed in the Kali routing configuration.

IP Address	Status	Observed Role	Assessment Relevance
192.168.64.1	Host up	Virtual network gateway	Network infrastructure asset
192.168.64.2	Host up	Kali Linux VM	Primary assessment asset

5.2 TCP Service Enumeration

The full TCP scan examined all 65,535 TCP ports on 192.168.64.2. Nmap reported all 65,535 ports as closed with TCP reset responses. No listening TCP services were identified.

5.3 Vulnerability Scan

The vulnerability-oriented Nmap scan used -sV and the vuln NSE script category against 192.168.64.2. This scan examined Nmap's default set of 1,000 TCP ports. All 1,000 scanned ports were reported as closed, so no exposed TCP service was available for the vulnerability scripts to assess. No vulnerability findings were reported by this scan.

5.4 UDP Scan

The UDP assessment examined the 100 most common UDP ports on 192.168.64.2. All 100 scanned ports were reported as closed with port-unreachable responses. No open UDP service was identified in the tested set.

6. Findings and Vulnerability Classification

No exploitable vulnerability was identified in the network services exposed by the Kali VM during this assessment. Because no vulnerability was actually detected, no artificial CVE, CWE, or OWASP vulnerability has been assigned. The primary security observation is an informational attack-surface finding.

ID	Finding	Evidence	Classification	Severity	Security Implication
F-001	No exposed TCP/UDP services detected in the tested ranges	Full TCP scan; top-100 UDP scan	Attack-surface / exposure assessment	Informational	A lack of exposed services reduces the remotely accessible attack

					surface of the VM.
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7. Security Implications

The Kali VM was reachable on the local virtual network, but no TCP services were listening across the full TCP port range tested.

The UDP scan also found no open services among the 100 most common UDP ports tested.

A minimal exposed service footprint generally reduces opportunities for remote network-based attacks.

The absence of detected vulnerabilities does not prove that the system is completely vulnerability-free. It indicates that no vulnerability was identified through the specific scans and port ranges documented in this report.

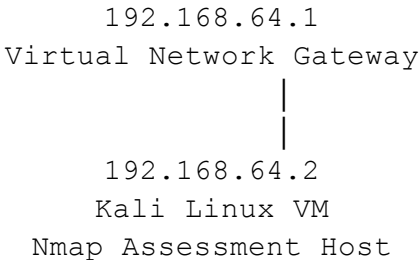
8. Limitations and Recommendations

The vulnerability-script command shown in the evidence used Nmap's default 1,000 TCP ports, while a separate service scan examined all 65,535 TCP ports. For completeness, the vulnerability-script scan could be repeated with `-p-` so that the vulnerability scripts are applied across the full TCP port range. In addition, the UDP assessment covered only the 100 most common UDP ports rather than all 65,535 UDP ports.

For a more comprehensive application-security demonstration, the project's intentionally vulnerable OWASP Juice Shop target can be assessed separately once it is reachable from Kali. That would provide application-level findings that are distinct from the network exposure results documented here.

9. Basic Network Map

Virtual Lab Network: 192.168.64.0/24



10. Evidence Screenshots

```

(root@kali)-[/home/kali]
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether ea:f0:2f:00:11:3a brd ff:ff:ff:ff:ff:ff
    inet 192.168.64.2/24 brd 192.168.64.255 scope global dynamic noprefixroute
        valid_lft 2831sec preferred_lft 2831sec
    inet6 fd1b:e2f0:f39e:1a01:d29c:6779:2177:9256/64 scope global temporary dynamic
        valid_lft 604034sec preferred_lft 85151sec
    inet6 fd1b:e2f0:f39e:1a01:e8f0:2fff:fe00:113a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591941sec preferred_lft 604741sec
    inet6 fe80::e8f0:2fff:fe00:113a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

```

(root@kali)-[/home/kali]
# nmap -sn 192.168.64.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-08 11:10 PDT
Nmap scan report for 192.168.64.1
Host is up (0.00041s latency).
MAC Address: DE:93:96:D0:90:64 (Unknown)
Nmap scan report for 192.168.64.2
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 1.93 seconds

```

```

(root@kali)-[/home/kali]
# nmap -sV -p- 192.168.64.2
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-08 11:26 PDT
Nmap scan report for 192.168.64.2
Host is up (0.00s latency).
All 65535 scanned ports on 192.168.64.2 are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds

```

```
(root@kali)-[/home/kali]
# nmap -sV --script vuln 192.168.64.2 -oN kali-vulnerability-scan.txt
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-08 11:30 PDT
Nmap scan report for 192.168.64.2
Host is up (0.0000030s latency).
All 1000 scanned ports on 192.168.64.2 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.32 seconds
```

```
(root@kali)-[/home/kali]
# nmap -sU --top-ports 100 192.168.64.2
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-09 08:32 PDT
Nmap scan report for 192.168.64.2
Host is up (0.0000010s latency).
All 100 scanned ports on 192.168.64.2 are in ignored states.
Not shown: 100 closed udp ports (port-unreach)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

11. Conclusion

The Nmap assessment successfully demonstrated asset discovery, TCP service enumeration, vulnerability-oriented scanning, and UDP service assessment against the Kali Linux VM. Two active hosts were identified on the virtual subnet, while the Kali VM itself showed no open TCP ports across all 65,535 ports and no open ports among the 100 UDP ports tested. The vulnerability scan produced no vulnerability findings. The results support the conclusion that the assessed Kali VM had a minimal network-exposed attack surface at the time of testing.